Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Sea Ice TAC
Arctic Sea Ice Age
SEAICE_ARC_PHY_SIAGE_L4_MY_011_025

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CHANGE RECORD

Issue	Date	§	Description of Change	Author	Validated By
1.0	November 2025	All	Creation of the document	A. Korosov	C. Wettre

ACRONYM TABLE

Acronym	Description
EAN	Estimated Accuracy Number
EASE	Equal-Area Scalable Earth
FYI	First-year ice
MYI	Multi-year ice
NRT	Near Real Time
OSI SAF	Ocean and Sea Ice Satellite Application Facilities
SIAge	Sea Ice Age
SIC	Sea ice concentration
SID	Sea Ice drift
TAC	Copernicus Marine Service Thematic Assembly Centre

KEY DEFINITIONS

Sea Ice Age

Intuitively, age is a measure of time, and, therefore, the sea ice age is the time since the beginning of ice freezing. However, it makes sense to consider age as a measure of time only when discussing individual floes of sea ice. A finite volume, which is represented by a pixel on satellite observations or an element in a model grid, always contains a mixture of floes with different ages. The proportion of floes with varying ages in this mixture evolves with time as ice freezes laterally, drifts and mixes with other ice, converges or diverges and refreezes from opened water in the leads and polynyas. Therefore, characterisation of the age of sea ice in a pixel (or model element) with one number is not sufficiently informative.

To overcome this problem, we provide a distribution of sea ice age fractions (see below) and a weighted average sea ice age in every pixel. Hereafter, this distribution is referred to as the “Sea Ice Age”.

Ice age fraction

Fraction of ice in a pixel with a given age category. The fraction is provided as a percentage of the total concentration. The age categories are provided with a precision of one year, counting from the start of the freezing season. Example age categories:

First-year ice (1YI): ice formed during the ongoing freezing season that has not yet experienced melting.

Second-year ice (2YI): ice formed during the previous freezing season and that has survived one melting season.

Example sea ice age information in a pixel of the Arctic Sea Ice Age product:

Category	1YI	2YI	3YI	4YI	5YI	6YI	7YI
Percentage	10%	25%	25%	20%	15%	10%	5%

Average sea ice age in the dataset

The average sea ice age is computed as a weighted average of age of ice fractions with weight equal ti the concentration of the ice fraction (see Equation 6).

Maximum sea ice age in the dataset

The maximum age is computed as the age of the fraction of ice covering a percentage of the total concentration within a pixel above 15%.

Age derived from the drifting buoy

The age derived from a drifting buoy equals the time since the buoy was deployed on sea ice. If the buoy ended up in open water during the melting season, the age of the buoy is computed as time since refreeze of the ice around the buoy

Accuracy

The accuracy of the product is computed as the ratio of correct age predictions (i.e., age from the product ≤ age from the buoy) to the total number of matchups.

Matchups

Collocated in space and time observations by the sea ice age dataset and the drifting buoys.

First-year ice

Ice formed during the ongoing freezing season that has not yet experienced melting.

Second-year ice

Ice formed during the previous freezing season that has survived one melting season.

Triangular mesh

A 2D triangular mesh is a collection of triangles connecting vertices (points) to discretize and represent a 2D surface or shape, commonly used in computer graphics and finite element analysis (FEA).

RELEVANT DOCUMENT LIST

CMEMS-SEAICE-PUM-011-025-v1.0	Product user manual for the Arctic Sea Ice Age, SEAICE_ARC_PHY_SIAGE_L4_MY_011_025
CMEMS-SEAICE-SQO-011-025-v1.0	Synthetic quality overview for the Arctic Sea Ice Age, SEAICE_ARC_PHY_SIAGE_L4_MY_011_025

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This QUID covers SEAICE_ARC_PHY_SIAGE_L4_MY_011_025, the Arctic Sea Ice Age, at 65 km resolution, spanning 1991 - 2024.

I.2 Summary of the results

The Arctic Sea Ice Age product is validated against trajectories of drifting sea ice buoys from 1991–2024 by comparing both average and maximum product-derived ice age with buoy-based ice age estimates. Buoy age is defined as the time since deployment, or since refreezing if the buoy entered open water during the melt season. Validation assumes the product age should be greater than or equal to the buoy age, and accuracy is quantified as the fraction of matchups where this condition is satisfied. The accuracy of the weighted average age is 86.3%, and the accuracy of the maximum age is 98.3%, see Table 1 (see further details below). The underestimation of the maximum ice age occurs most often at the ice edge, where the concentration and drift fields are the most uncertain.

I.3 Estimated Accuracy Numbers

Estimated Accuracy Numbers (EANs) for the Arctic Sea Ice Age product are provided below.

Table 1. Table of estimated accuracy numbers (where variables, units and numbers are given).

Variable	Units	Metrics	Value of metric
Average age	Years	Accuracy	86.3%
Maximum age			98.3%

II PRODUCTION SYSTEM DESCRIPTION

Here we describe the multi-year product Arctic Sea Ice Age (SIAge) produced by the Nansen Environmental and Remote Sensing Centre in Bergen, Norway. It spans the period from 1991 through 2024. The dataset is based on a novel Lagrangian advection scheme applied to a triangular mesh, which conserves sea ice age fractions and reduces numerical diffusion compared to the previous Eulerian approach (Korosov et al., 2026). SIAge is derived from satellite observations of sea ice concentration and drift, and represents fractional age classes per grid cell. The record captures the spatial and temporal evolution of first- to sixth-year ice, including uncertainty estimates that account for both sea ice concentration and drift.

The estimates are computed in the following way. A pan-Arctic field of concentration is taken from satellite observations at the end of the melt season, when the ice extent is minimal, and all ice is assumed to be multi-year ice (MYI, C_{MY}). The C_{MY} field is repetitively advected (morphed) using daily satellite-observed sea ice drift fields. Changes of concentration due to ice deformation or melting are accounted for during the advection process. Some time after the initialization of the MYI field advection, the advected field represents concentration of MYI, which is lower than the total observed concentration and is denoted as C_{A0} (i.e., advected for less than one year). The difference between the total and advected fields yields the concentration of the first-year ice:

$$C_{1Y} = C_{TOT} - C_{A0}. \quad \text{(Equation 1)}$$

Hereafter, we define the first-year ice (C_{1Y}) as ice formed during the ongoing freezing season and that has not yet experienced melting.

One year after the initialisation, the advected field represents the concentration of sea ice which is at least two years old and is denoted C_{A1} (advected for one year). At that time, the total observed concentration again reaches a minimum, representing the concentration of multi-year ice. CMY has higher values than the advected C_{A1} as it also contains a fraction of the second-year ice:

$$C_{2Y} = C_{MY} - C_{A1}. \quad \text{(Equation 2)}$$

Hereafter, we define the second-year ice (C_{2Y}) as ice formed during the previous freezing season and that has survived one melting season.

Both C_{A1} and the current C_{MY} are advected further using the sea ice drift and after one more year they become C_{A2} and C_{A1}, i.e., ice fractions advected for two years and for one year. Since C_{A1} contains C_{A2} and, similar to the previous year, the new C_{MY} contains C_{A1} we can compute fractions of the second- and third-year ice:

$$C_{2Y} = C_{MY} - C_{A1} \quad \text{(Equation 3)}$$

$$C_{3Y} = C_{A1} - C_{A2} \quad \text{(Equation 4)}$$

This workflow is repeated, and the ice age fraction can be computed using a generic formula:

$$C_{NY} = C_{A(N-2)} - C_{A(N-1)}, \quad \text{(Equation 5)}$$

where N is an integer number indicating the age of sea ice fraction.

In this algorithm, we use a Lagrangian scheme of advection of MYI ice, where nodes of a triangular mesh are advected using the ice drift vectors. Triangular elements in this mesh are advected without diffusion unless a remeshing occurs. A detailed description of the input data, advection, remeshing and validation is provided in Korosov et al., 2025.

The algorithm for mesh advection was initialised on 1 January 1991 and advected the mesh using Sea Ice Drift (SID) Climate Data Record (CDR) until 2020, and using the (Near Real Time) NRT SID product thereafter. The MYI concentration field was computed for 15 September for years 1991 – 2024. Each MYI concentration field is advected for 2200 days (six years). For the spin-up period before 15 September 1995, we had fewer fields with advected MYI thus, for each day after 15 September 1995, we obtained six fields of C_{A1}–C_{A5}. Since the same meshes and mappings were used for advection of MYI concentration originating from different years, for each day, the C_{A1}–C_{A5} fields are located on the same mesh, which facilitates computation of sea ice age fractions, see Eqs. 1 - 5. For dates between the minimum ice extent, the first-year ice concentration was also computed.

The sea ice age is then computed using a weighted average as suggested in our previous algorithm:

$$A_{AVG} = \Sigma_i A_i C_{iY} / \Sigma_i C_{iY} \qquad \text{(Equation 6)}$$

where Ai ∈ [1,2,3,4,5,6,7] is an integer year, C_{iY} is the fraction of ice of corresponding age (C_{1Y} corresponds to FYI).

Values of the ice age fractions, as well as the average ice age, were linearly interpolated on a regular grid in the EASE2 (Equal-Area Scalable Earth Grid) projection with a resolution of 65 km and exported to netCDF files.

Figures 1 and 2 show an example of the content of the sea ice age product for 1st January 2000.

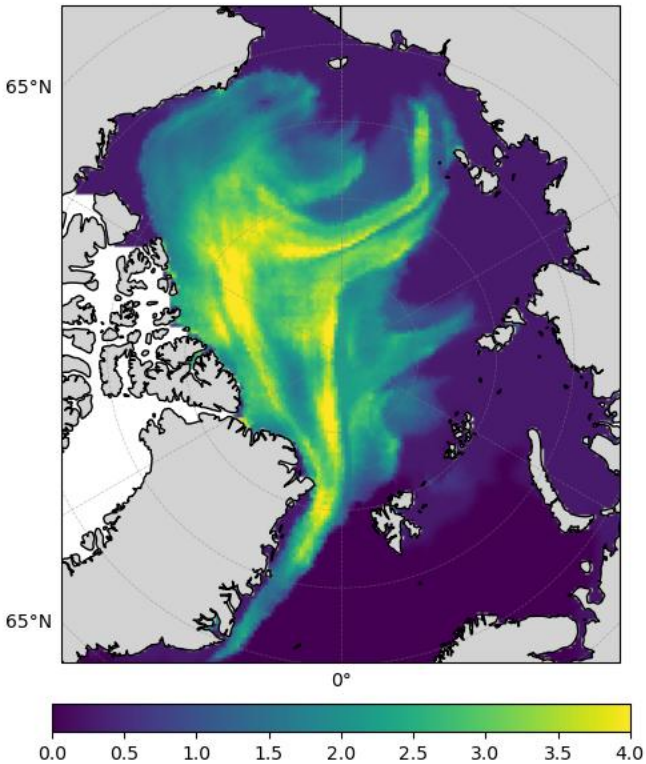

Figure 1. *Example content of sea ice age file for 1 January 2000. The file contains the weighted average sea ice age shown in this figure and the concentrations of seven ice age fractions (see Fig. 2).*

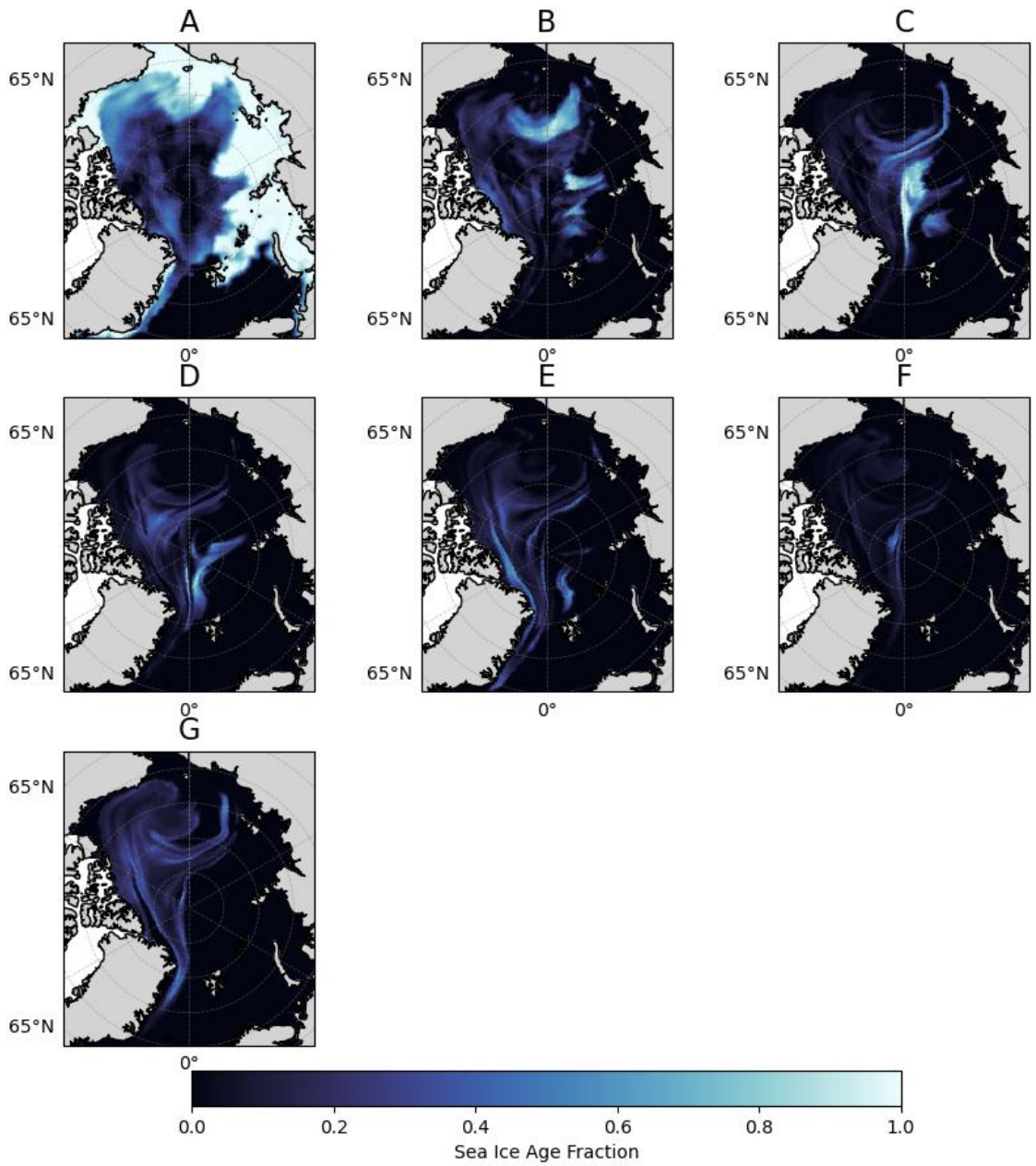

***Figure 2**. Example content of sea ice age file for 1 January 2000. The file contains the weighted average of sea ice age (see Fig. 1) and the concentrations of seven ice age fractions, as shown in the figure. A - fraction of the first-year ice. B - fraction of the second-year ice. C - fraction of the third-year ice. D - fraction of the fourth-year ice. E - fraction of the fifth-year ice. F - fraction of the sixth-year ice. G - fraction of the seventh-year ice.*

III VALIDATION FRAMEWORK

This section describes the framework that was followed to validate the product.

The Arctic Sea Ice Age product is validated using trajectories of sea ice drifting buoys during the period 1991 – 2024. The average sea ice age and the maximum sea ice age from the product are compared to the age obtained from the drifting buoy. Age from the drifting buoy is computed as time since deployment, or, if the buoy ended up in open water during the melting season, since refreeze of the ice around the buoy. The comparison is performed under the expectation that the age of the product is larger than or equal to the age of the drifting buoy. The accuracy "A" of the product is computed as the ratio of correct age predictions (i.e., age from the product <= age of the buoy) to the total number of matchups.

For validation, we used the same reference data as in (Lavergne and Down, 2023) acquired between 1991 and 2024. Figure 3 shows the spatial distribution of the number of ice drifting buoys as well as the spatial distribution of the average age of the buoys.

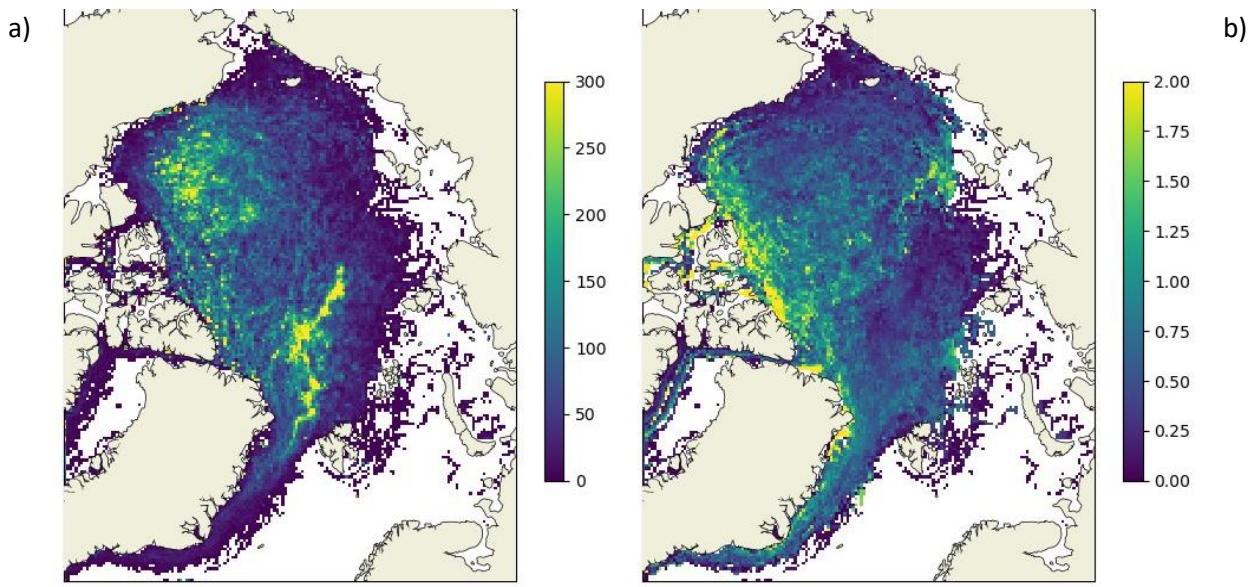

Figure 3. *Number of buoy positions (a) collected between 1991 and 2024, and average age of the buoys (b). The map's geographic location in this figure matches that in Figures 1 and 3.*

III.1 Validation using sea ice drifting buoys

We compare the average and maximum ice age detected by the Lagrangian algorithm to the age of the buoys. These two quantities are not expected to match one-to-one because we do not know the age of the ice on which the buoy was deployed. Instead, the validation tests the hypothesis that the age of the buoy cannot exceed the maximum ice age (see the Key Definitions) detected by our SIAge product. The age of the ice-drifting buoy is calculated from the time of its deployment. The buoy trajectories are collocated with the OSI SAF SIC product (Lavergne et al., 2019) to detect the melting and freezing of the ice around the buoy. If a buoy is deployed in open water or ends up in open water, then OSI SAF concentration drops below 15 %. During summer it melts and then refreezes again into sea ice thus the concentration grows above 15 % and the age counting is reset to 0.

IV VALIDATION RESULTS

This section provides the results derived from the validation of the Arctic Sea Ice Age dataset using the sea ice drifting buoys.

IV.1 Validation using sea ice drifting buoys

The comparison of the Arctic Sea Ice Age product with the ice drifting buoys shows that in the majority of cases, the ice age obtained from a buoy is lower than the collocated average or maximum ice age, which is as desired (Figure 4a and 4b). In only 1.7 % of cases is the buoy age older (i.e., there are errors in the estimates provided by the product) than the maximum age of the Arctic Sea Ice Age product. The map in Figure 4c shows that the underestimation of the maximum ice age occurs most often at the ice edge, where the concentration and drift fields are the most uncertain.

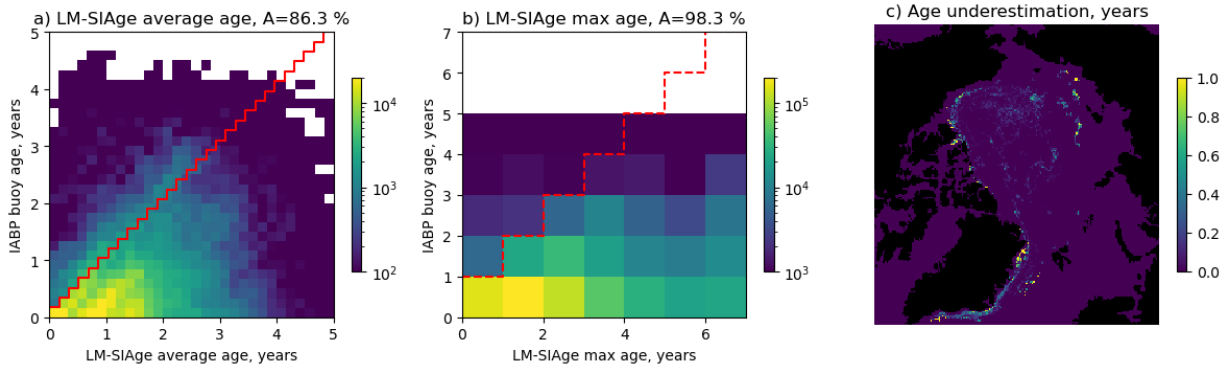

***Figure 4**. Comparison of the Arctic Sea Ice Age product with the age derived from the collocated ice drifting buoys. a) Arctic Sea Ice Age weighted average age, b) Arctic Sea Ice Age maximum age. c) Map of the ice age is overestimation in the Arctic Sea Ice Age dataset compared to the buoys, along with the magnitude of the discrepancy. The red line on (a) and (b) shows the separation between correct (below the line) and incorrect (above the line) predictions. Accuracy (A) of each product is provided in the title. Colours denote the number of matchups.*

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system. It includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among other changes that may impact product quality.

No changes, outages of noticeable events were noticed.

Date	Change/Event description	System version	Other

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version.

This section does not apply to the present version of the product.

VII REFERENCES

Korosov, A., Edel, L., Regan, H., Lavergne, T., Down, E. J., and Aaboe, S.: A Climate Data Record of Sea Ice Age Using Lagrangian Advection of a Triangular Mesh, Earth Syst. Sci. Data, https://doi.org/10.5194/essd-18-721-2026, 2026.

Lavergne, T., Sørensen, A. M., Kern, S., Tonboe, R., Notz, D., Aaboe, S., Bell, L., Dybkjær, G., Eastwood, S., Gabarro, C., Heygster, G., Killie, M. A., Brandt Kreiner, M., Lavelle, J., Saldo, R., Sandven, S., and Pedersen, L. T.: Version 2 of the EUMETSAT OSI SAF and ESA CCI sea-ice concentration climate data records, The Cryosphere, 13, 49–78, https://doi.org/10.5194/tc-13-49-2019, 2019.

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